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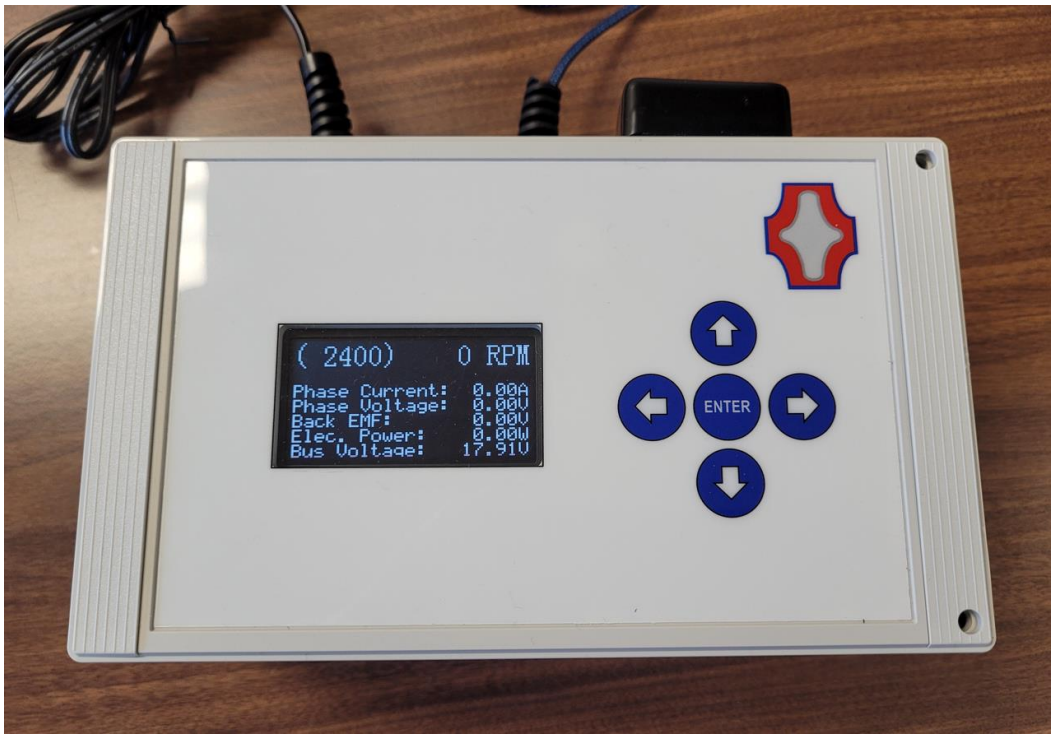
RBTS Controller Instructions

May 04, 2023 Rev 03

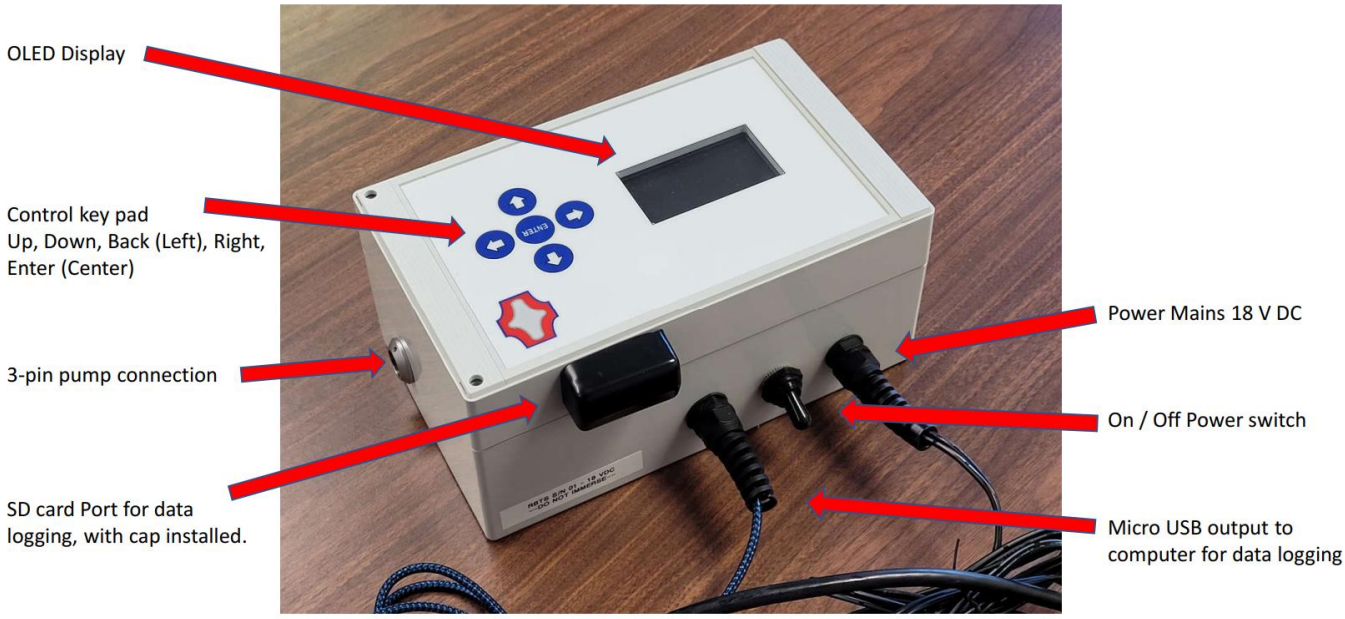


SPECIFICATIONS

- Minimum speed – 1000 RPM
- Max speed – Current limited ~ 4000 to 4500 RPM
- Max board current output – 1500 mA
- Auto-tuning R & L for attached motor
- USB data logging
- SD Data logging – hot swappable
- 18 VDC power supply
- Onboard date and time clock with battery

CAUTIONS

- Do not immerse controller in liquid
- Do not reverse wire the power supply POS and NEG
- When installing SD card be sure to insert into card holder
- It is preferred to utilize minimum current to start motor. This startup current is limited by the Max Current Setting
- Do not run pump dry. Sliding surface bearings must be submerged in fluid.



SCREENS

Note: Values will auto scroll with a long button press.

a. TOP LEVEL

i. Target speed (RPM)
ii. Measured speed (RPM)
iii. Phase Current, Peak – Phase to neutral
iv. Phase Voltage, Peak – Phase to neutral
v. Back EMF Peak – Phase to neutral
vi. Electrical Power to motor (W)
vii. Power Supply / Bus Voltage - (V DC)

b. MOTOR SPEED

i. Enable; ENTER – Starts motor
ii. Disable; ENTER – Stops Motor
iii. Note – Speed may be limited by max current delivered from controller board. If Measured Speed is lagging Target Speed double check the Max Current setting and increase if possible
iv. For EMERGENCY STOP either switch toggle switch to OFF or unplug power supply.

c. MAX CURRENT

i. Max board output is 1500 mA
ii. Recommended startup current ~ 1200 mA. Once started increase to 1500 mA. If Pump does not start, apply more current at startup by increasing Max Current.

d. LOGGING FREQUENCY

i. Enter desired USB and SD card logging frequency from 1 Hz to 100Hz.
ii. See data map below.

e. AUTO TUNE

i. Controller will automatically sweep sine frequencies through motor to determine system resistance and inductance.
ii. RECOMMENDATION IS TO AUTO TUNE AT OPERATIONAL TEMPERATURE
iii. Auto tune cycle takes about 2 minutes to complete.
iv. Determined data values are latched in memory

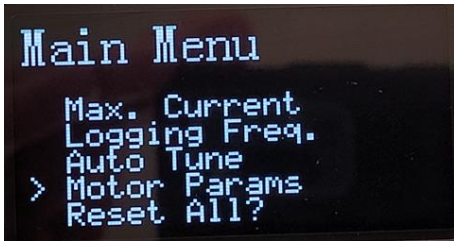
f. MOTOR PARAMETERS

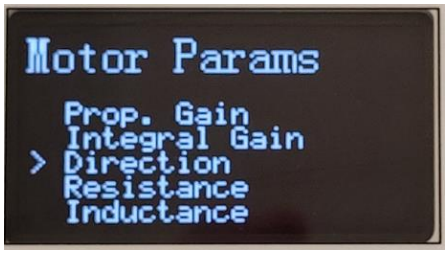
i. Proportional Gain – Factory set.
   1. Recommended range for KP is 5 to 15.
ii. Integral Gain – Factory set.
   1. Recommended value for KI is 10.
iii. Direction
   1. Forward - Pump should spin in counterclockwise direction when viewed from IVC end
   2. Reverse
iv. Resistance – Measured Phase to Neutral resistance (milli-Ohm).
   1. Resistance is set by the autotuner and should not be changed.
v. Inductance – Measured Phase to Neutral Inductance (micro-Henry).
   1. Inductance is set by the autotuner and should not be changed.
vi. Identifier – Software entered controller serial number. This should match label on PCB and enclosure. S/N is logged on USB and SD card datasets.

g. FACTORY RESET

Motor Direction

During test setup, user should ascertain that pump is spinning in the correct direction (CCW when viewed from the IVC – wire end of pump). The correct rotation will rotate the impeller vanes in correct direction to pump fluid in the direction of the tangentially oriented PA outlets. If Pump does not spin in the correct direction, follow the steps below to change the motor rotational direction.





MOTOR PARAMS – ENTER.
DIRECTION – ENTER.
Toggle Direction option to REVERSE.
ENTER.

SD CARD DATA MAP

Operational data displayed on controller front panel is logged to a .CSV file at the desired logging frequency.

Notes: First line of data file may be nonsensical, errata – just ignore or delete.

0	0	0.002	0	0	0	17.91	*ERRATA - IGNORE THIS LINE*		
DATE	**TIME**	**CONTROLLER S/N**	**SET POINT SPEED (RPM)**	**SPEED** MEASURED (RPM)	**CURRENT** PHASE TO NEUTRAL (A PEAK)	**VOLTAGE** PHASE TO NEUTRAL (V PEAK)	**BEMF (V PEAK** PHASE TO NEUTRAL)	**ELEC. POWER TO MOTOR (W)**	**POWER SUPPLY** BUS VOLTAGE (VDC)
3/10/2023	11:49:09.668	1	2400	2403	0.547	3.63	2.06	2.97	17.94
3/10/2023	11:49:09.718	1	2400	2402	0.548	3.63	2.06	2.98	17.94
3/10/2023	11:49:09.768	1	2400	2403	0.548	3.63	2.06	2.98	17.94
3/10/2023	11:49:09.818	1	2400	2403	0.548	3.63	2.06	2.98	17.94
3/10/2023	11:49:09.867	1	2400	2401	0.55	3.63	2.06	2.99	17.94
3/10/2023	11:49:09.917	1	2400	2404	0.547	3.62	2.06	2.97	17.95
3/10/2023	11:49:09.967	1	2400	2401	0.549	3.63	2.06	2.99	17.94
3/10/2023	11:49:10.017	1	2400	2403	0.55	3.63	2.06	3	17.94
3/10/2023	11:49:10.067	1	2400	2403	0.548	3.63	2.06	2.98	17.94
3/10/2023	11:49:10.117	1	2400	2403	0.547	3.62	2.06	2.97	17.94
3/10/2023	11:49:10.167	1	2400	2402	0.551	3.63	2.06	3	17.94
3/10/2023	11:49:10.217	1	2400	2406	0.545	3.62	2.06	2.96	17.94
3/10/2023	11:49:10.267	1	2400	2401	0.55	3.63	2.06	2.99	17.94

TECHNICAL NOTES

1. Formula for the PI controller: phase current = KP * ((velocity error) + KI * integral (velocity error)), with phase current in milliamps and velocity error in mechanical rad/s.
2. Because the phase current is delivered exactly in phase with the back emf mechanical power can be estimated as 1.5 * (back emf) * (phase current). This mechanical power estimate does not include eddy current losses: eddy currents can be modeled as a viscous drag torque proportional to velocity.
3. Displayed electrical power figure is strictly real power (no reactive component included)
4. Phase voltage is not perfectly aligned with phase current, that is current lags behind voltage because of the motor's inductance. Multiplying them with a 1.5x scale factor will give reactive power as well as real power into the motor. V * I * 1.5 = Total Power

Appendix A – Time sync Instructions.

From: Aaron Yeiser
Sent: Friday, March 10, 2023 9:05 PM
Subject: Python time sync code

Installation:

Install python 3.11: https://www.python.org/downloads/

Open cmd

Run this command:

pip install pyserial datetime

Run:

Connect the USB cable to the computer. Open device manager to determine port. Replace /dev/ttyACM0 or COM4 with the port that the motor controller shows up as.

Linux:

python send_time.py /dev/ttyACM0

Windows:

python send_time.py COM4

This will synchronize the motor controller to computer time to within a few ms, although there is measurable clock drift.

The USB port will emit some garbage timestamp data for about a second after synchronization--this is an artifact of the STM32's clock hardware and how it handles time synchronization.

Appendix B – Software Revision 01, 18-APR-2023.

New features and modifications:

1) Display SD CARD MOUNTED when inserting SD card. Alternate main menu screen was added that display the detection when an SD card is inserted. New file name is displayed when SD card is detected. File name is comprised of a Date and Time string.
2) SYSTEM TIME –
    a. Display current system date and time at startup for 10 sec. This will allow user to verify if system time is correct or it needs to be corrected or annotated.
    b. Add in a menu option to see the system time and date
    c. Make hardware correction to oscillator circuit was implemented for better time/date accuracy.
3) Control motor acceleration rate at user main-menu inputted speed changes
    a. User can input motor acceleration ramp rate in the range of 100 rpm/sec to 900 rpm/sec.
4) More time was allotted for user to make a change before it reverts back to the default screen. Increased to 15 secs.